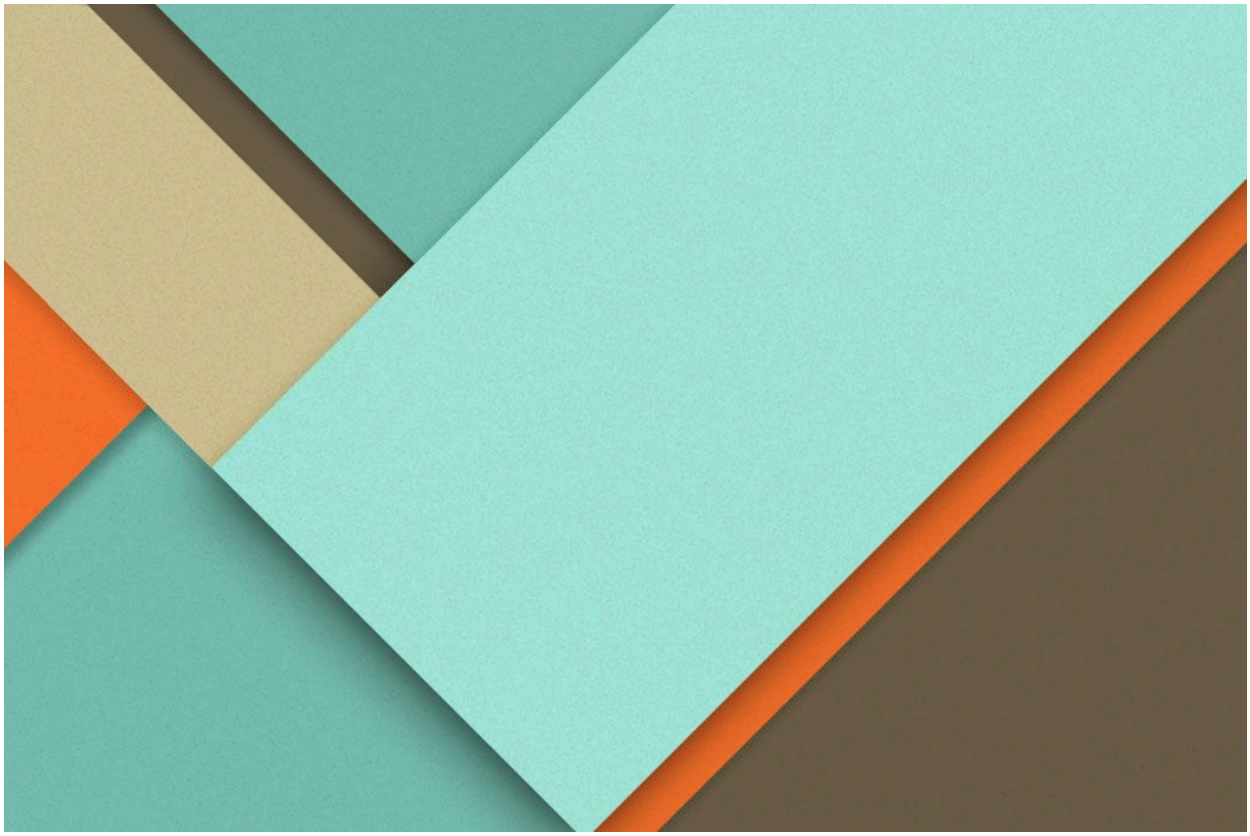


Optimity Access ltd.
Documentation and best practices for high-performance infrastructure migration, cloud modernization, server optimization, and scalable system architecture.

Specifications

Comprehensive Operational & Architectural Checklist for Zero-Loss, High-Performance Infrastructure Transition

1. Business & Migration Planning

☐ Define migration objectives (performance, scalability, security, cost optimization)
☐ Identify critical business services and dependencies
☐ Define acceptable downtime and SLA requirements
☐ Prepare migration timeline and milestones
☐ Assign migration roles and responsibilities
☐ Establish risk assessment and mitigation plan
☐ Define rollback strategy before starting migration

---

2. Current Infrastructure Assessment

☐ Document existing architecture and system components
☐ Analyze current servers, databases, networks, and applications
☐ Identify legacy systems and technical limitations
☐ Map all application dependencies
☐ Review third-party integrations and external services
☐ Analyze current CPU, RAM, storage, and network usage
☐ Identify performance bottlenecks

---

3. Target Architecture Design

☐ Design new infrastructure architecture
☐ Select suitable server environment:

- Cloud
- Dedicated server
- Hybrid infrastructure

☐ Define application architecture:

- Monolith
- Microservices
- Serverless

☐ Plan database architecture
☐ Design API communication strategy

☐ Define caching strategy (Redis/CDN)
☐ Plan load balancing architecture
☐ Design high availability and failover mechanisms

---

4. Server Infrastructure Preparation

☐ Provision new servers and environments
☐ Configure operating systems and dependencies
☐ Install required runtime environments
☐ Configure web servers (Nginx/Apache)
☐ Configure application servers
☐ Setup storage systems
☐ Configure networking and security groups
☐ Enable monitoring agents

---

5. Database Migration Strategy

☐ Create complete database backup
☐ Verify backup integrity
☐ Analyze database size and structure
☐ Plan schema migration
☐ Optimize database indexes and queries
☐ Test data migration process
☐ Perform data validation after migration
☐ Enable replication if required
☐ Ensure zero data loss during transition

---

6. Data Integrity & Validation

☐ Compare old and new database records
☐ Verify user accounts and permissions

☐ Validate transaction history
☐ Check file and media storage integrity
☐ Verify timestamps and relationships
☐ Run automated data consistency checks
☐ Maintain migration audit logs

---

7. Application Migration

☐ Review application compatibility
☐ Update environment configurations
☐ Migrate application code
☐ Update dependencies and libraries
☐ Configure environment variables
☐ Update API endpoints
☐ Verify authentication systems
☐ Test application functionality

---

8. Performance Optimization

☐ Optimize server resources
☐ Configure caching layers
☐ Optimize database queries
☐ Enable CDN where required
☐ Compress static assets
☐ Optimize API response time
☐ Configure connection pooling
☐ Perform load testing

---

9. Security Hardening

- ☐ Configure firewall rules
- ☐ Enable SSL/TLS certificates
- ☐ Implement encryption standards
- ☐ Secure database access
- ☐ Configure identity and access management
- ☐ Enable vulnerability scanning
- ☐ Review security permissions
- ☐ Perform penetration testing

---

10. Network & Communication Setup

- ☐ Configure DNS migration plan
- ☐ Verify domain routing
- ☐ Configure API gateways
- ☐ Test internal network communication
- ☐ Optimize latency between services
- ☐ Configure VPN/private networking
- ☐ Validate external service connectivity

---

11. Testing & Quality Assurance

- ☐ Perform unit testing
- ☐ Perform integration testing
- ☐ Perform API testing
- ☐ Perform security testing
- ☐ Perform performance testing
- ☐ Perform stress and load testing
- ☐ Test failure scenarios
- ☐ Validate user workflows

---

12. Deployment Strategy

☐ Select deployment method:

- Blue-Green Deployment
- Rolling Deployment
- Canary Deployment

☐ Prepare production deployment checklist
☐ Freeze unnecessary changes before migration
☐ Perform final backup
☐ Execute migration steps according to plan
☐ Monitor system during deployment

---

13. Zero-Downtime Migration Process

☐ Synchronize old and new environments
☐ Enable real-time data replication
☐ Redirect traffic gradually
☐ Monitor errors during transition
☐ Keep old infrastructure available temporarily
☐ Confirm successful traffic migration
☐ Decommission old systems only after validation

---

14. Monitoring & Observability

☐ Setup server monitoring
☐ Monitor CPU, RAM, disk, and network usage
☐ Configure application logging
☐ Setup error tracking
☐ Monitor database performance
☐ Track API response time

☐ Configure alert notifications
☐ Create operational dashboards

---

15. Backup & Disaster Recovery

☐ Maintain automated backups
☐ Test backup restoration
☐ Define Recovery Point Objective (RPO)
☐ Define Recovery Time Objective (RTO)
☐ Maintain disaster recovery environment
☐ Document recovery procedures

---

16. User Experience Validation

☐ Verify website/application speed
☐ Test mobile and desktop experience
☐ Validate login and account functions
☐ Test payment and transaction flows
☐ Monitor user feedback
☐ Check regional performance

---

17. Cost Optimization

☐ Review server resource utilization
☐ Remove unnecessary resources
☐ Optimize cloud services
☐ Review storage costs
☐ Optimize database expenses
☐ Monitor monthly infrastructure spending

---

18. Documentation & Knowledge Transfer

☐ Document new architecture
☐ Document server configuration
☐ Document deployment process
☐ Document database structure
☐ Document security policies
☐ Train operation and development teams
☐ Maintain troubleshooting guides

---

19. Post-Migration Operations

☐ Monitor system stability
☐ Analyze performance improvements
☐ Fix migration-related issues
☐ Optimize based on real usage data
☐ Review security status
☐ Update disaster recovery plans

---

20. Final Migration Approval Checklist

☐ All data verified successfully
☐ No critical errors detected
☐ Performance targets achieved
☐ Security requirements satisfied
☐ Backup confirmed
☐ Monitoring active

☐ Rollback plan tested
☐ Business team approved migration completion

Final Goal:

Zero Data Loss + Minimal Downtime + Improved Performance + Higher Scalability + Strong Security + Future Expansion Capability

Made by
Sohag Hossain
IT Team